

Learning ML

A practical guide to foundation models, datasets, fine-tuning, job status, inference, and comparison — written for product users and technical teams.

1. Foundation model

- What it is: A foundation model is a large, pre-trained model that already understands broad patterns from massive amounts of data.
- What it does: It gives you a strong starting point, so you do not need to train a model from scratch.
- When to use it: Use it when you want faster time to value, less training data, and better baseline performance.
- Example: A language model that can summarize text, answer questions, and classify messages after light adaptation.

2. Dataset

- What it is: The dataset is the labeled or unlabeled information used to adapt and evaluate the model.
- What it does: It teaches the model your domain, your labels, and your business rules.
- When to use it: Use a high-quality dataset when accuracy matters and your use case is specific.
- Example: Customer support tickets labeled as billing, technical issue, or cancellation.

3. Launch fine-tune

- What it is: Fine-tuning updates a foundation model using your dataset so it performs better on your task.
- What it does: It adapts the model to your language, categories, policies, or visual patterns.
- When to use it: Use it when the base model is close, but not good enough for your production target.
- Example: Fine-tuning a classifier to detect internal support categories.

4. Job status

- What it is: Job status shows the state of the training or fine-tuning run.
- What it does: It tells users whether the job is queued, running, completed, failed, or stopped.
- When to use it: Use it for monitoring progress, troubleshooting failures, and estimating completion time.
- Example: A training job running for 28 minutes with logs updating every few seconds.

5. Inference

- What it is: Inference is the act of using the trained model to make a prediction on new input.
- What it does: It returns a label, score, probability, or generated response.
- When to use it: Use it whenever the application needs a live prediction from the trained model.
- Example: Classifying a new email as urgent or non-urgent.

6. Compare

- What it is: Compare lets users evaluate runs, models, or datasets side by side.
- What it does: It helps identify the best model based on metrics, cost, latency, and quality.
- When to use it: Use it before promoting a model to production.
- Example: Comparing version A and version B on accuracy, recall, and latency.

Key concepts

Concept	Meaning
Overfitting	Great on training data, weak on new data.
Underfitting	Too simple to learn the pattern well.
Ideal fit	Strong training and validation performance with small gap.
Accuracy	Percent of correct predictions.
Precision	How many positive predictions were correct.
Recall	How many real positives were found.
Loss	How wrong the model is during training.

How to read model quality

- High accuracy is useful, but it is not enough by itself. A model can be accurate overall and still miss the important class.
- Low accuracy usually means the model needs better data, better labels, a better base model, or more tuning.
- Look at accuracy together with precision, recall, F1 score, validation loss, and inference latency.

Recommended visuals for the page

Visual	Purpose
Training vs validation loss curve	Shows underfitting, overfitting, and ideal fit.
Accuracy bar chart	Compares model versions or runs.
Confusion matrix	Shows which classes are being confused.
Job status timeline	Explains training progress.
Latency vs accuracy plot	Highlights tradeoffs for inference.

Suggested UI behavior

- Place the Learning ML link in the left menu.
- Open this content in the right panel as a scrollable article.
- Add section anchors or tabs for quick navigation.
- Use callout cards and charts to keep the page engaging and easy to scan.